

Yurou Liu

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RESEARCH INTEREST

LLM4Science — building large language models and generative foundation models that speak the language of science, spanning AI for Science, generative foundation models, molecular representation learning, 3D molecular pre-training, and reinforcement learning.

EDUCATION

Renmin University of China

Ph.D. Student in Artificial Intelligence (Advisor: Prof. Bing Su)

Beijing

Sept. 2023 – Present

Beijing Normal University

B.Sc. in Computer Science

Beijing

Sept. 2019 – June 2023

INTERNSHIP EXPERIENCE

Tongyi, Alibaba

Research Intern on LLM4Science (Mentor: Mingyang Li)

Beijing

Dec. 2024 – Present

PUBLICATIONS

* denotes equal / core contribution.

Speaking the Language of Science: Toward a General-Purpose Generative Foundation Model for the Natural Sciences

Mingyang Li*, Yurou Liu*, Jieping Ye, Bing Su, Ji-Rong Wen, Zheng Wang

Tongyi Tech Report 2026

DrugTrail: Interpretable Drug Discovery via Structured Reasoning and Druggability-Tailored Preference Optimization

Yurou Liu, Mingyang Li, Xinyuan Zhu, Rui Jiao, Yiming Dong, Xinyu Tang, Yang Liu, Jieping Ye, Bing Su, Zheng Wang

ICLR 2026

Towards High Data Efficiency in Reinforcement Learning with Verifiable Reward

Xinyu Tang*, Zhenduo Zhang*, Yurou Liu*, Wayne Xin Zhao, Zujie Wen, Zhiqiang Zhang, Jun Zhou

ICLR 2026

DenoiseVAE: Learning Molecule-Adaptive Noise Distributions for Denoising-based 3D Molecular Pre-training

Yurou Liu*, Jiahao Chen*, Rui Jiao*, Jiangmeng Li, Wenbing Huang, Bing Su

ICLR 2025

Probing RLVR Training Instability through the Lens of Objective-Level Hacking

Yiming Dong, Kun Fu, Haoyu Li, Xinyuan Zhu, Yurou Liu, Lijing Shao, Jieping Ye, Zheng Wang

ICML 2026

Learning 3D Anisotropic Noise Distributions Improves Molecular Force Field Modeling

NeurIPS 2025

Xixian Liu, Rui Jiao, Zhiyuan Liu, **Yurou Liu**, Yang Liu,
Ziheng Lu, Wenbing Huang, Yang Zhang, Yixin Cao